



**NORTH
SYDNEY
GIRLS HIGH
SCHOOL**

2024

**HSC
Assessment
Task 3**

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 60 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Section II, show relevant mathematical reasoning and/or calculations

Total marks – 40

Section I – 4 marks (pages 2 – 3)

- Attempt Questions 1 – 4
- Allow about 7 minutes for this section

Section II – 36 marks (pages 5 – 7)

- Attempt Questions 5-7
- Allow about 53 minutes for this section

NAME: _____

TEACHER: _____

STUDENT NUMBER:

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Question	1 – 4	5	6	7	Total
Inverse Functions	Q1,4 /2	/3	/4	/5	/14
Further Integration Techniques	Q2,3 /2	/3	/8		/13
Further Trigonometry and Rates of Change		/6		/7	/13
	/4	/12	/12	/12	/40

Section I

4 marks

Attempt Questions 1-4

Allow about 7 minutes for this section

Use the multiple choice answer sheet for Questions 1-4.

1 What is the domain of the function $\sin^{-1}(3x)$?

A. $-\frac{\pi}{6} \leq x \leq \frac{\pi}{6}$

B. $-\frac{1}{3} \leq x \leq \frac{1}{3}$

C. $-\frac{3\pi}{2} \leq x \leq \frac{3\pi}{2}$

D. $-3 \leq x \leq 3$

2 Which of the following is a primitive of $\cos^2\left(\frac{2x}{3}\right)$?

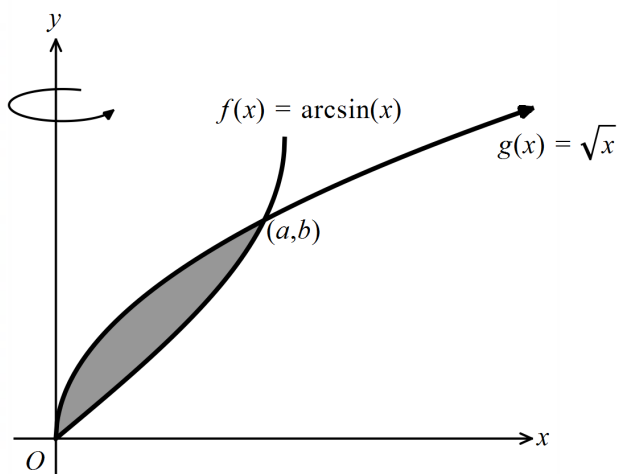
A. $\frac{x}{2} + \frac{3}{4} \sin\left(\frac{4x}{3}\right)$

B. $\frac{x}{2} - \frac{3}{4} \sin\left(\frac{4x}{3}\right)$

C. $\frac{x}{2} + \frac{3}{8} \sin\left(\frac{4x}{3}\right)$

D. $\frac{x}{2} - \frac{3}{8} \sin\left(\frac{4x}{3}\right)$

- 3 The curves $f(x) = \arcsin x$ and $g(x) = \sqrt{x}$ intersect at $(0,0)$ and (a,b) in the first quadrant as shown. The region enclosed by the curves is rotated about the y -axis.



Which of the following gives the volume of the solid of revolution formed?

- A. $\pi \int_0^a x - (\sin^{-1} x)^2 dx$
- B. $\pi \int_0^b \sin^2 y - y^2 dy$
- C. $\pi \int_0^b (\sin y - y^2)^2 dy$
- D. $\pi \int_0^b \sin^2 y - y^4 dy$
- 4 What is the exact value of $\sin^{-1}(\cos(a\pi))$ where a can be any value in the interval $(1, 1.5)$?
- A. $a\pi$
- B. $\pi - a\pi$
- C. $a\pi - \frac{3\pi}{2}$
- D. $a\pi + \frac{\pi}{2}$

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Section II

36 marks

Attempt Questions 5-7

Allow about 53 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In questions 5-7, your responses should include relevant mathematical reasoning and/or calculations.

Question 5 (12 marks) Use a SEPARATE writing booklet

- (a) Differentiate $\cos^{-1}(3x)$. **1**
- (b) Find $\int_0^{\frac{\sqrt{3}}{2}} \frac{dx}{9+4x^2}$. **2**
- (c) Use the substitution $u = 3x+1$ to evaluate $\int_0^8 \frac{x}{\sqrt{3x+1}} dx$ **3**
- (d) The radius of a circular oil spill is increasing at a rate of 3 metres per minute. **2**
Find the rate at which the area covered by the oil spill is increasing, when the radius is 20 metres?
- (e) By writing $\cos x - \sin x$ in the form $R \cos(x + \alpha)$, where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$, **4**
solve $\cos x = \sin x + 1$ in the domain $0 \leq x \leq 2\pi$.

Question 6 (12 marks) Use a SEPARATE writing booklet

(a) Find $\int \cos 3\theta \sin 5\theta \, d\theta$. 2

(b) Let $f(x) = x^3 + 3x + 1$.

(i) Explain why $f^{-1}(x)$ exists. 1

(ii) Find the point(s) on the graph of $y = f^{-1}(x)$ where the gradient is $\frac{1}{6}$. 3

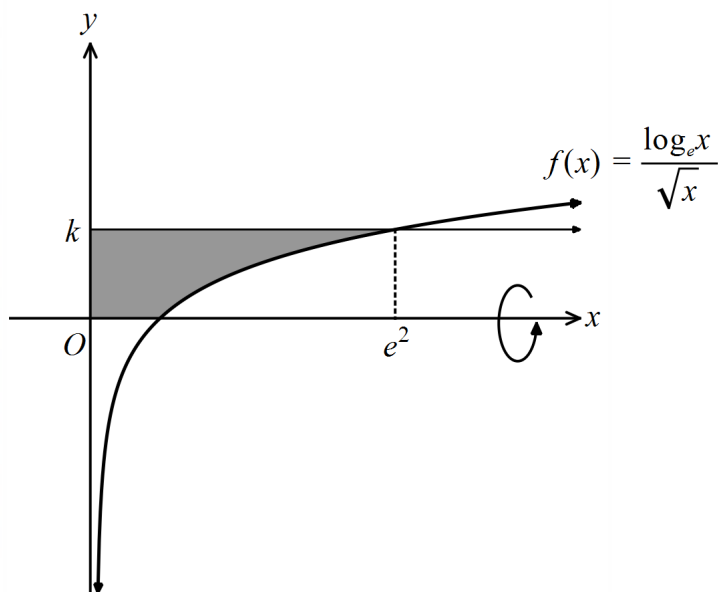
(c) (i) Use the substitution $u = \log_e x$ to find $\int \frac{(\log_e x)^2}{x} dx$. 2

(ii) Consider the function $f(x) = \frac{\log_e x}{\sqrt{x}}$. 4

The line $y = k$ intersects the graph of $y = f(x)$ at $x = e^2$.

A student designs a glass ornament which can be modelled by rotating the region bounded by the curve, the coordinate axes and the line $y = k$ about the x -axis.

By first finding the value of k , find the exact volume of glass used in the ornament.



Question 7 (12 marks) Use a SEPARATE writing booklet

- (a) You are given that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$. DO NOT PROVE THIS.
- (i) By substituting $x = \cos \theta$ in the polynomial equation $8x^3 - 6x + 1 = 0$, show that $\cos 3\theta = -\frac{1}{2}$. 2
- (ii) Using the result of part (i), prove that $\cos \frac{2\pi}{9} \cos \frac{4\pi}{9} \cos \frac{8\pi}{9} = -\frac{1}{8}$. 2
- (b) Consider $f(x) = \tan^{-1}\left(\frac{\sin x}{\cos x - 1}\right)$, $0 < x < \pi$.
- (i) Show that $f'(x)$ is constant. 3
- (ii) Hence sketch the graph of $f(x)$. 2
- (c) A curve C is defined by the following set of parametric equations: 3

$$\left. \begin{array}{l} x = \sin \theta \\ y = e^{-\cos \theta} \end{array} \right\}, \text{ where } 0 \leq \theta \leq 2\pi$$

A point P traces the curve C as the value of θ changes at a constant rate of 0.2 radians per second.

Find the exact rate at which the gradient m of the curve at P is changing when $\theta = \frac{2\pi}{3}$.

NOTE: The gradient m of the curve C at any point P on the curve is given by $\frac{dy}{dx}$.

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Multiple Choice

1. $-1 \leq 3x \leq 1$

$$-\frac{1}{3} \leq x \leq \frac{1}{3}$$

$\therefore$ (B)

2.
$$\begin{aligned}\int \cos^2 \frac{2x}{3} dx &= \frac{1}{2} \int (1 + \cos \frac{4x}{3}) dx \\ &= \frac{1}{2} (x + \frac{3}{4} \sin \frac{4x}{3}) \\ &= \frac{x}{2} + \frac{3}{8} \sin \frac{4x}{3}\end{aligned}$$

$\therefore$ (C)

3. For outer radius

$$y = \arcsin x$$

$$x = \sin y$$

$$V_{\text{outer}} = \pi \int_0^b \sin^2 y dy$$

For inner radius

$$y = \sqrt{x}$$

$$x = y^2$$

$$V_{\text{inner}} = \pi \int_0^b y^4 dy$$

$$V = \pi \int_0^b (\sin^2 y - y^4) dy$$

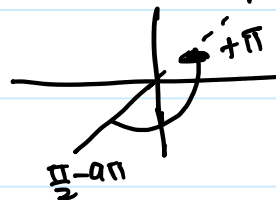
$\therefore$ (D)

4. $\sin^{-1}(\cos a\pi) = \sin^{-1}(\sin(\frac{\pi}{2} - a\pi))$

Now $\pi < a\pi < \frac{3\pi}{2}$

$\therefore 1 < a < \frac{3}{2}$

$$-\pi < \frac{\pi}{2} - a\pi < -\frac{\pi}{2}$$



Related angle = $(\frac{\pi}{2} - a\pi) + \pi$
in 1st Quad
 $= \frac{3\pi}{2} - a\pi$

$$\sin^{-1}(\sin(\frac{\pi}{2} - a\pi)) = -(\frac{3\pi}{2} - a\pi) = a\pi - \frac{3\pi}{2}$$

$\therefore$ (C)

Question 5

$$a) \quad \frac{d}{dx} \cos^{-1}(3x) = \frac{-1}{\sqrt{1-(3x)^2}} \times 3 = \frac{-3}{\sqrt{1-9x^2}}$$

$$\begin{aligned} b) \quad \int_0^{\frac{\sqrt{3}}{2}} \frac{dx}{9+4x^2} &= \frac{1}{2} \int_0^{\frac{\sqrt{3}}{2}} \frac{dx}{3^2 + (2x)^2} \\ &= \frac{1}{2} \times \frac{1}{3} \left[\tan^{-1} \frac{2x}{3} \right]_0^{\frac{\sqrt{3}}{2}} \\ &= \frac{1}{6} \left(\tan^{-1} \left(\frac{1}{\sqrt{3}} \right) - \tan^{-1} 0 \right) \\ &= \frac{1}{6} \times \left(\frac{\pi}{6} - 0 \right) \\ &= \frac{\pi}{36} \end{aligned}$$

$$\begin{aligned} (c) \quad \int_0^8 \frac{x}{\sqrt{3x+1}} dx & \quad \left| \quad \begin{array}{l} \text{Let } u = 3x+1 \\ \frac{du}{dx} = 3 \quad \text{so} \quad \frac{du}{3} = dx \end{array} \right. \\ = \int_1^{25} \frac{u-1}{3\sqrt{u}} \times \frac{du}{3} & \quad \left| \quad \begin{array}{l} \text{Also, } x = \frac{u-1}{3} \end{array} \right. \\ = \frac{1}{9} \int (u-1) u^{-\frac{1}{2}} du & \quad \left| \quad \begin{array}{c|c|c} x & 0 & 8 \\ \hline u & 1 & 25 \end{array} \right. \\ = \frac{1}{9} \int u^{\frac{1}{2}} - u^{-\frac{1}{2}} du & \\ = \frac{1}{9} \left[\frac{2}{3} u^{\frac{3}{2}} - 2 u^{\frac{1}{2}} \right]_1^{25} &= \frac{1}{9} \left[\left(\frac{2}{3} \times 125 - 10 \right) - \left(\frac{2}{3} - 2 \right) \right] \\ &= \frac{224}{27} \end{aligned}$$

Question 5 (contd)

(d) $\frac{dr}{dt} = 3$ $\frac{dA}{dt} = ?$ when $r = 20$

$$A = \pi r^2$$

$$\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt} \quad (\text{Chain Rule})$$

$$= (2\pi r) \cdot 3$$

$$= 6\pi r$$

$$\text{When } r = 20, \quad \frac{dA}{dt} = 6\pi \times 20 \\ = 120\pi$$

The area covered by the oil spill is increasing at a rate of $120\pi \text{ m}^2/\text{min}$.

(e) Consider $\cos x - \sin x = R \cos(x + \alpha)$

$$= R \cos \alpha \cos x - R \sin \alpha \sin x$$

Equating coefficients of $\cos x$ and $\sin x$

$$R \cos \alpha = 1 \quad (1)$$

$$R \sin \alpha = 1 \quad (2)$$

$$(1)^2 + (2)^2 : \quad R^2 (\cos^2 \alpha + \sin^2 \alpha) = 1^2 + 1^2$$

$$R^2 = 2$$

$$R = \sqrt{2}, \quad R > 0$$

$$\text{From (1)} \quad \cos \alpha = \frac{1}{R} = \frac{1}{\sqrt{2}}$$

$$\therefore \alpha = \frac{\pi}{4} \quad \text{as } 0 \leq \alpha \leq \frac{\pi}{2}$$

Question 5 contd

(e) contd.

$$\therefore \cos x - \sin x = \sqrt{2} \cos\left(x + \frac{\pi}{4}\right)$$

$$\text{Now } \cos x = \sin x + 1$$

$$\text{so } \cos x - \sin x = 1$$

$$\therefore \sqrt{2} \cos\left(x + \frac{\pi}{4}\right) = 1$$

$$\cos\left(x + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \quad 0 \leq x \leq 2\pi$$

$$\frac{\pi}{4} \leq x + \frac{\pi}{4} \leq 2\pi + \frac{\pi}{4}$$

$$x + \frac{\pi}{4} = \frac{\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}$$

$$x = 0, \frac{3\pi}{2}, 2\pi$$

Question 6

$$\begin{aligned} \text{a) } \int \cos 3\theta \sin 5\theta \, d\theta &= \int \frac{\sin(5\theta + 3\theta) + \sin(5\theta - 3\theta)}{2} \, d\theta \\ &\text{using products to sums } A=5\theta \quad B=3\theta \\ &= \frac{1}{2} \int (\sin 8\theta + \sin 2\theta) \, d\theta \\ &= \frac{1}{2} \left(-\frac{\cos 8\theta}{8} - \frac{\cos 2\theta}{2} \right) + C \\ &= -\frac{\cos 8\theta}{16} - \frac{\cos 2\theta}{4} + C \\ &= -\frac{1}{16} (\cos 8\theta + 4\cos 2\theta) + C \end{aligned}$$

$$\text{(b) } f(x) = x^3 + 3x + 1$$

$$\text{(i) } f'(x) = 3x^2 + 3$$

$$> 0 \quad \text{for all } x \quad \text{as } x^2 \geq 0$$

$\therefore f(x)$ is monotonically increasing and

$\therefore f^{-1}(x)$ exists

(ii) corresponding points on $f(x)$ have a gradient of 6.

$$f'(x) = 6 \Rightarrow 3x^2 + 3 = 6$$

$$3x^2 = 3$$

$$x^2 = 1$$

$$x = \pm 1$$

Pts on $f(x)$ with gradient of 6 are $(1, 5)$ and $(-1, -3)$

Corresponding pts on $f(x)$ $(5, 1)$ and $(-3, -1)$

have a gradient of $\frac{1}{6}$.

Question 6 contd

$$\begin{array}{l} \text{(c) (i)} \quad \int \frac{(\ln x)^2}{x} dx \\ \quad = \int u^2 du \\ \quad = \frac{u^3}{3} + C \end{array} \quad \left| \quad \begin{array}{l} \text{Let } u = \ln x \\ \frac{du}{dx} = \frac{1}{x} \text{ so } du = \frac{1}{x} dx \end{array} \right.$$

$$= \frac{(\ln x)^3}{3} + C$$

$$\text{(ii)} \quad k = \frac{\ln(e^2)}{\sqrt{e^2}} = \frac{2}{e}$$

$$\text{Volume} = \text{Volume}_{\text{cylinder}} - \text{Volume}_{\text{solid of revolution}}$$

$$= \pi \left(\frac{2}{e}\right)^2 \times e^2 - \pi \int_1^{e^2} \left(\frac{\ln x}{\sqrt{x}}\right)^2 dx$$

$$= \pi \times \frac{4}{\cancel{e^2}} \times \cancel{e^2} - \pi \int_1^{e^2} \frac{(\ln x)^2}{x} dx$$

$$= 4\pi - \pi \left[\frac{(\ln x)^3}{3} \right]_1^{e^2} \quad \text{using (i)}$$

$$= 4\pi - \pi \left(\frac{8}{3} - 0 \right)$$

$$= 4\pi - \frac{8\pi}{3}$$

$$= \frac{4\pi}{3}$$

$\therefore$ volume of glass used is $\frac{4\pi}{3} \text{ cm}^3$.

Question 7

(a) $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ (given)

(i) $8x^3 - 6x + 1 = 0$ Sub $x = \cos\theta$

$$8\cos^3\theta - 6\cos\theta + 1 = 0$$

$$2(4\cos^3\theta - 3\cos\theta) = -1$$

$$2\cos 3\theta = -1$$

$$\cos 3\theta = -\frac{1}{2}$$

(ii) $\cos 3\theta = -\frac{1}{2}$

$$3\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{12\pi}{3}, \dots$$

$$\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}, \frac{12\pi}{9}, \dots$$

As $x = \cos\theta$ then $x = \cos\frac{2\pi}{9}, \cos\frac{4\pi}{9}, \cos\frac{8\pi}{9}$

are the solutions to $8x^3 - 6x + 1 = 0$

Using product of roots:

$$\cos\frac{2\pi}{9} \cdot \cos\frac{4\pi}{9} \cdot \cos\frac{8\pi}{9} = -\frac{1}{8} \quad \left(-\frac{d}{a}\right)$$

(b) $f(x) = \tan^{-1}\left(\frac{\sin x}{\cos x - 1}\right)$

$$f'(x) = \frac{1}{1 + \left(\frac{\sin x}{\cos x - 1}\right)^2} \times \frac{\cos x (\cos x - 1) - \sin x (-\sin x)}{(\cos x - 1)^2}$$

$$= \frac{(\cancel{\cos x} - 1)^2}{(\cos x - 1)^2 + \sin^2 x} \times \frac{\cos^2 x - \cancel{\cos x} + \sin^2 x}{(\cancel{\cos x} - 1)^2}$$

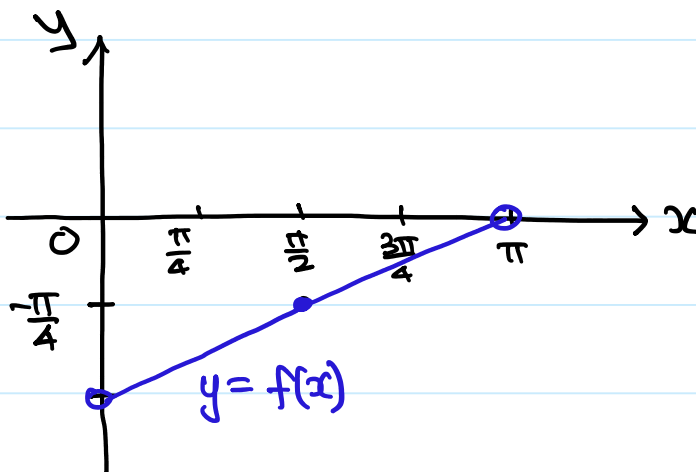
Question 7 (contd)

(b) contd.

$$\begin{aligned}f'(x) &= \frac{1 - \cos x}{\cos^2 x + 1 - 2\cos x + \sin^2 x} \\&= \frac{1 - \cos x}{2 - 2\cos x} \\&= \frac{1 - \cancel{\cos x}}{2(1 - \cancel{\cos x})} \\&= \frac{1}{2}\end{aligned}$$

(ii) $f'(x) = \frac{1}{2}$ so graph of $f(x)$ is a straight line with gradient $\frac{1}{2}$.

$$\begin{aligned}\text{Now } f\left(\frac{\pi}{2}\right) &= \tan^{-1}\left(\frac{1}{0-1}\right) \\&= \tan^{-1}(-1) \\&= -\frac{\pi}{4}\end{aligned}$$



Question 7 (contd)

(c) $x = \sin \theta$ $\frac{d\theta}{dt} = 0.2$ (given)
 $y = e^{-\cos \theta}$

$$\frac{dm}{dt} = ? \text{ when } \theta = \frac{2\pi}{3}$$

$$m = \frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx} \quad (\text{Chain Rule})$$

$$= \frac{dy}{d\theta} \times \frac{1}{\frac{dx}{d\theta}}$$

$$= \sin \theta e^{-\cos \theta} \times \frac{1}{\cos \theta}$$

$$= \tan \theta e^{-\cos \theta}$$

$$\text{So, } \frac{dm}{dt} = \frac{dm}{d\theta} \times \frac{d\theta}{dt} \quad (\text{Chain Rule})$$

$$= \left[\sec^2 \theta (e^{-\cos \theta}) + \tan \theta \cdot \sin \theta (e^{-\cos \theta}) \right] \times 0.2$$

$$\begin{aligned} \text{At } \theta = \frac{2\pi}{3}, \quad \frac{dm}{dt} &= \left((-2)^2 e^{+\frac{1}{2}} + (-\sqrt{3}) \frac{\sqrt{3}}{2} e^{+\frac{1}{2}} \right) \times \frac{1}{5} \\ &= \sqrt{e} \times \left(4 - \frac{3}{2} \right) \times \frac{1}{5} \\ &= \frac{1}{2} \sqrt{e} \end{aligned}$$

The gradient of the curve at P is changing at a rate of $\frac{\sqrt{e}}{2}$ per second.